

Math 241, F14
Quiz 6

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

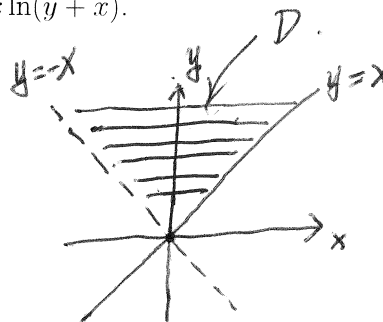
Problem 1. (4 points) Find and sketch the domain of the function

$$f(x, y) = \sqrt{y-x} \ln(y+x).$$

$$\begin{cases} y-x \geq 0 \\ y+x > 0 \end{cases}$$

$$\Rightarrow \begin{cases} y \geq x \\ y > -x \end{cases}$$

$$D = \{(x, y) \mid y \geq x \text{ and } y > -x\}$$



Problem 2. (6 points) Find the limit if it exists or show that the limit does not exist.

(a) $\lim_{(x,y) \rightarrow (1,-1)} e^{-xy} \cos(x+y)$

$$= e^{-(1) \cdot (-1)} \cos(1+(-1))$$

$$= e \cos 0$$

$$= e$$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}$

When $(x, y) \rightarrow (0, 0)$ along the x -axis,

$$f(x, y) = \frac{0^4}{x^4 + 3 \cdot 0^4} = 0$$

When $(x, y) \rightarrow (0, 0)$ along the y -axis,

$$f(x, y) = \frac{y^4}{0^4 + 3y^4} = \frac{1}{3}$$

Thus the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}$

does not exist.